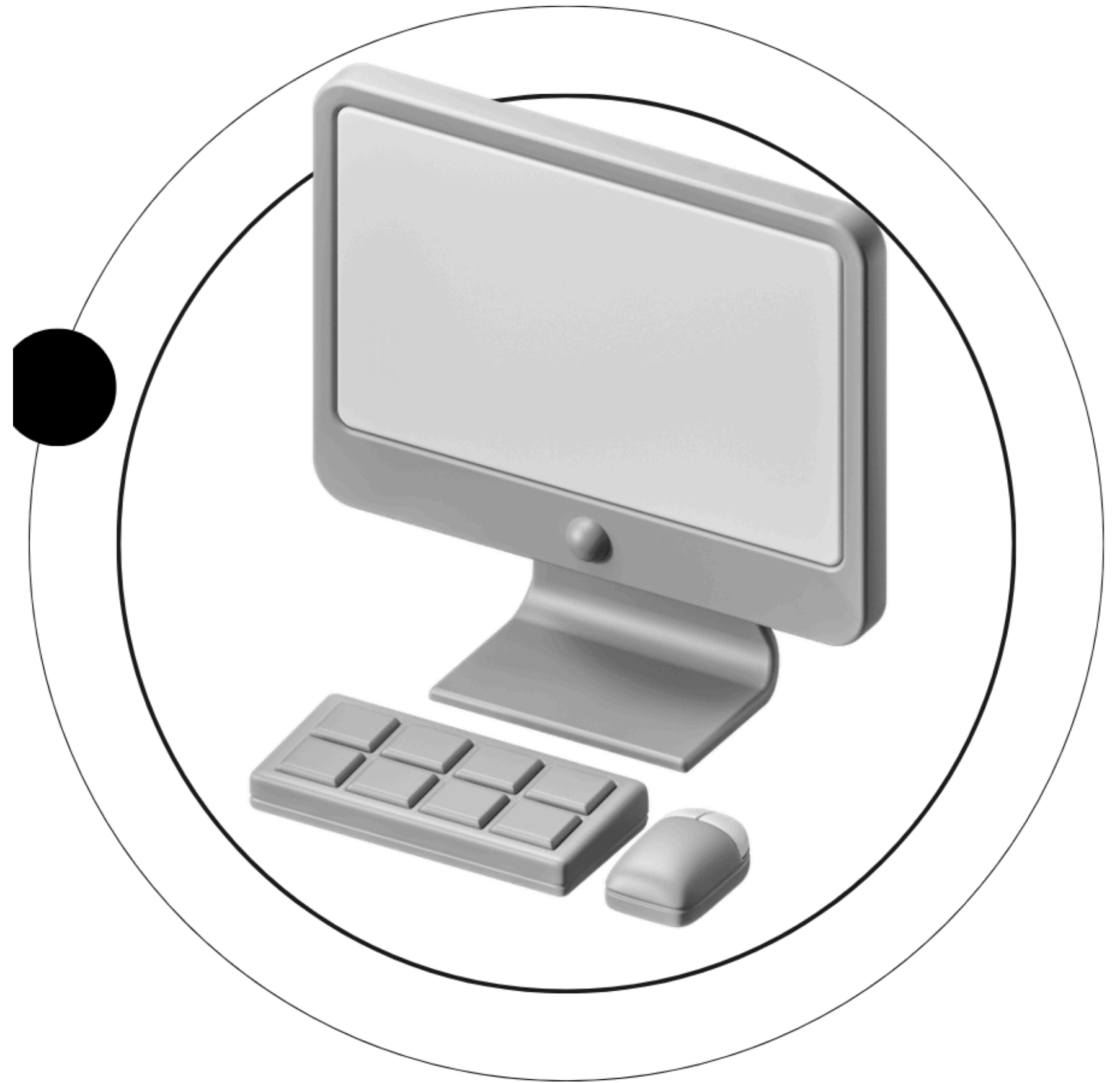


FOOD RECOGNITION AND CALORIE ESTIMATION

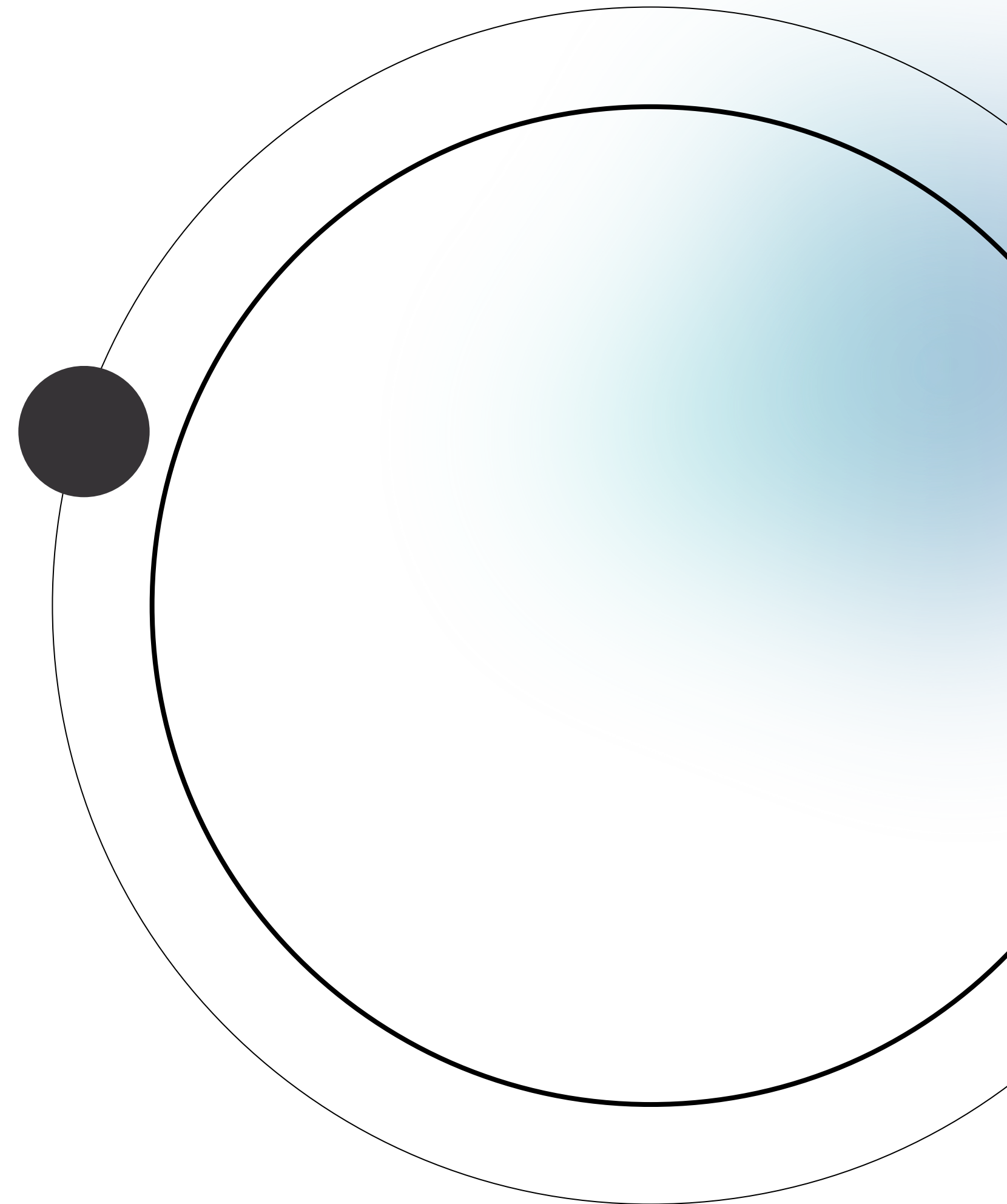
ABSTRACT: SYSTEM THAT IDENTIFIES FOOD ITEMS AND ESTIMATES THEIR CALORIC CONTENT FROM IMAGES. USES DEEP LEARNING FOR FOOD RECOGNITION AND NUTRITIONAL ANALYSIS."

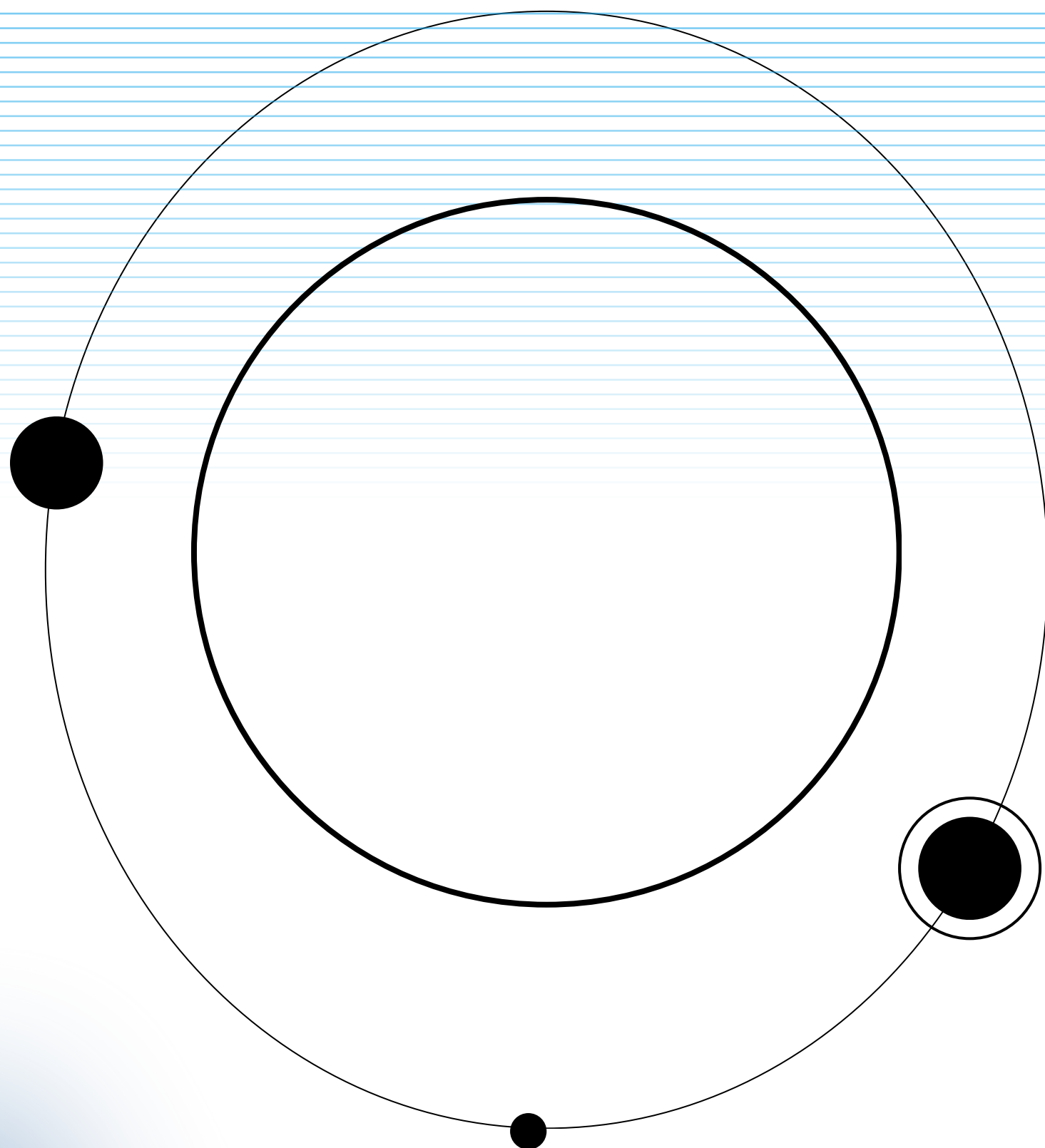


TEAM DETAILS

TEAM MEMBERS

1. Srija Lukka - SE22UCSE315
2. Shreyas Aditya Baksi - SE22UCSE317
3. Varshini Babers- SE22UCSE320
4. Harsha B - SE22UCSE321





INTRODUCTION

Our project was on Food Recognition and Calorie Estimation. The objectives of our model were to:

- Develop a model capable of predicting the calorie content of food items with high precision.
- Use CNN and ViT models to classify food items from categorized images
- Assess the models' performance and compare the metrics of both of the algorithms

METHODOLOGY

Our plan to precisely identify the number of calories by taking food image input is:

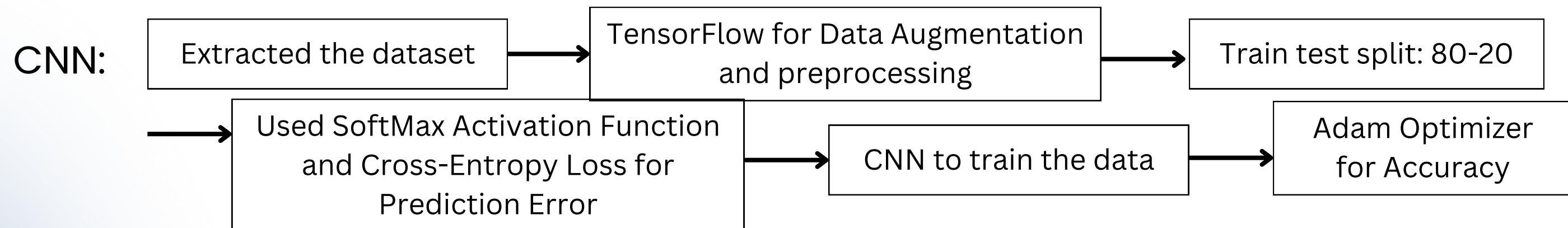
Data Collection

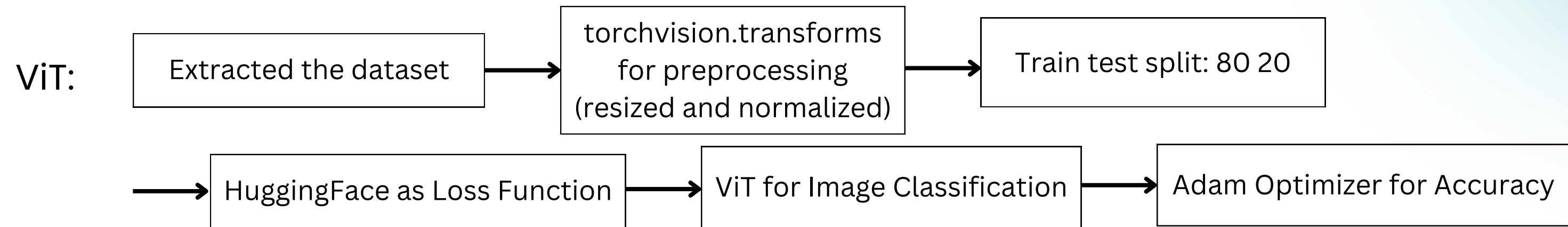
- Collected large amount of data to train the model against food images.
 - Split the data into train and test sets to evaluate performance of the model
 - Ensured that most of the data obtained was already categorized well, and appropriate for processing
-

Selection of Model

- CNN Model: Implemented a Convolutional Neural Network (CNN) for feature extraction and classification of food items.
- Purpose for choosing **CNN**: Can detect local patterns (like edges, textures, and shapes). Ideal for identifying features in food images.
- ViT Model: Utilized a Vision Transformer (ViT) model for capturing global patterns and achieving better performance on complex image features.
- Purpose for choosing **ViT**: ViT is capable of accurately assessing the performance of a model with complex images.

Implementation





Comparison Between Models

Accuracy: CNN – 97%, ViT – 50%

Applications: CNN – Complex; has real world applications, ViT – Handles lightweight tasks

Challenges:

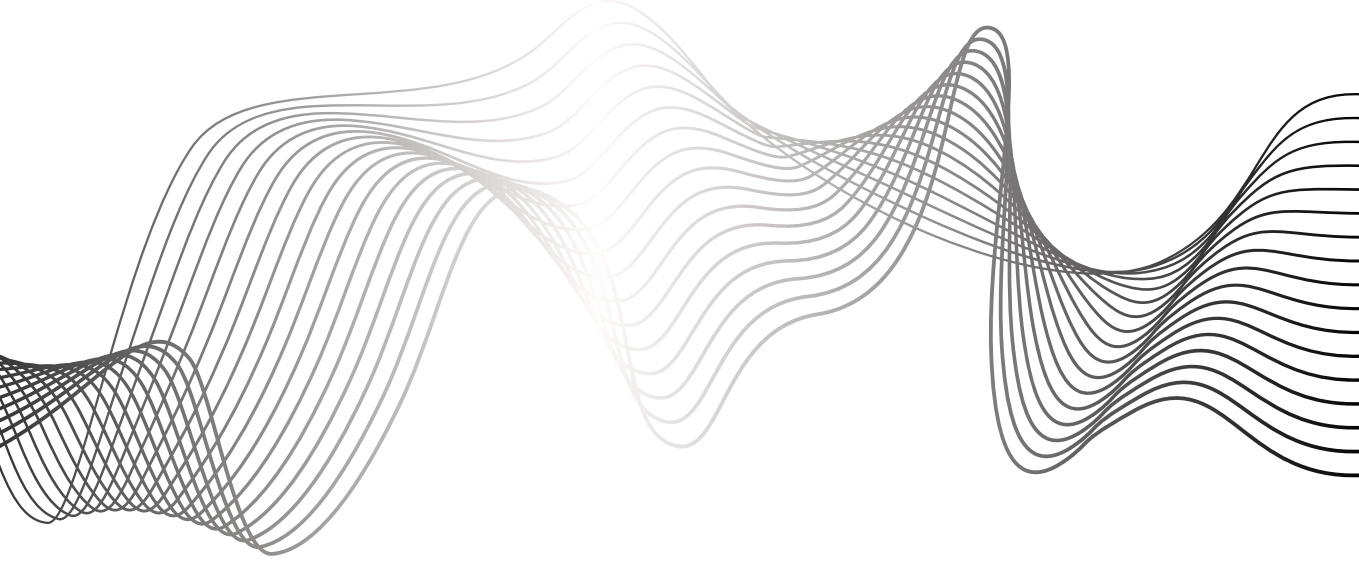
CNN – Data could have been made more diverse. There was some overfitting due to recurring similarity in images

ViT – Struggled with variations in terms of angles and overlapping items. ViT requires a very large dataset due to the high number of parameters.

Architecture:

CNN: Uses a hierarchical feature extraction approach. Learns complex features like shapes or objects in deeper layers.

ViT: Uses a Transformer architecture. Originally made for processing sequences like text in NLP, it works on images without using spatial convolutions.



RESULTS

CNN-

https://colab.research.google.com/drive/1myqug7xu6Om_gGgP_ZTRH-UwlqoVq6SUx?authuser=1#scrollTo=FNbOzlfSJdst

ViT-

https://colab.research.google.com/drive/1I24kwUIP-q_Wme7c8mAkwjRliGOnn1Om?authuser=2

Report-

<https://docs.google.com/document/d/1WCZJXAZFkosD9DqV0xMMIfPjDnIHMO3byWVoZ2AY9fw/edit?tab=t.0>

Dip Codes PDF-

<https://drive.google.com/drive/folders/1QYa9TklzyyeInC8giaEaADZgOdxF6o0B>

CNN Accuracy – 97% ViT Accuracy – 50%

CONCLUSIONS



● LIMITATIONS

The datasets we used had **limited examples of foods** which were tough to accurately predict.

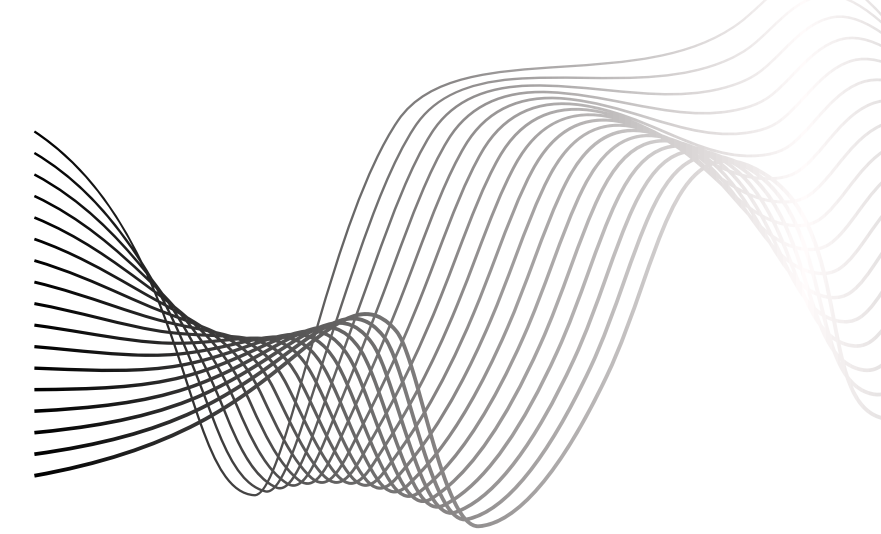
There were some **outlier overlapping objects** which decreased the accuracy for the model.

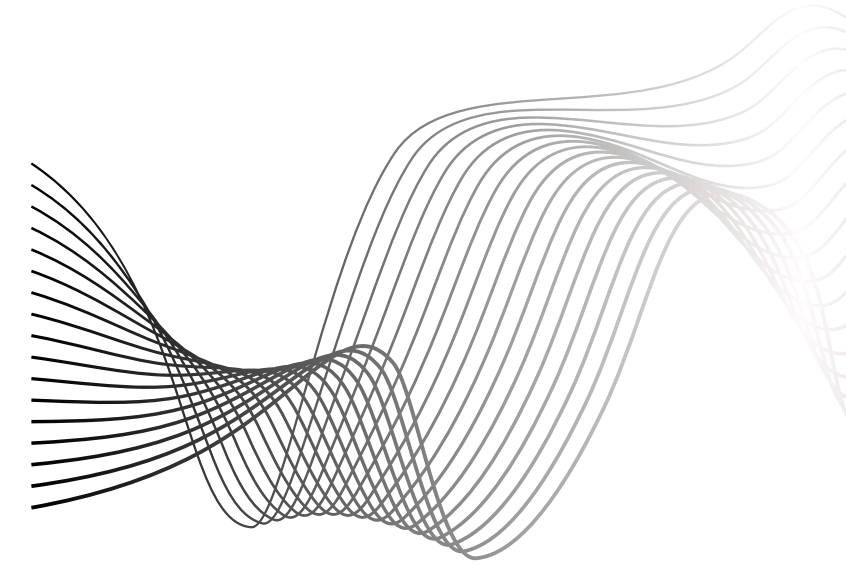
● INSIGHTS / IMPLICATIONS

ViT: There were few peaks and dips in the **eval/samples_per_second** and **eval/steps_per_second** metrics, which implies a small inconsistency in data, the data was not diverse enough

The graphs in **eval/loss** and **eval/runtime** were flat, which could imply that the model was underfitting and missed some patterns in the data. Hardware performance was consistent

Accuracy was **50%** for ViT. Implies that ViT was underfitting the data and overcomplicated certain simple data in cases.





CNN: Accuracy is **0.97** for CNN. This could suggest that there is some overfitting in the training of the model.

The training loss is **0.087** by the 10th Epoch, which suggests that the model was fitting the training data well.

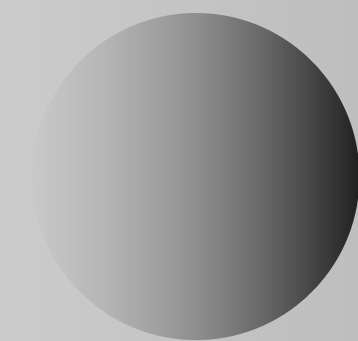
The validation accuracy is **0.9324** and the validation loss was minimal as well - **0.1446**

The accuracies were consistently **tending to 1**, which implies that the model is correctly predicting the majority of the outputs in the given dataset.

● RESULTS

- **CNN:** Achieved **97%** accuracy for basic food item classification. Struggled with overlapping and some image copies in data. The model was underfit on the data
- **ViT:** Achieved **50%** accuracy for basic food item classification. Caloric estimations were consistent and precise. The model was overfit on the data





THANK YOU!

